# 3. Software process

"Software Development Methodologies"
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## Process - Motivation

**Software development process** The set of tasks, the sequence and flow of these tasks, the inputs to and the outputs from the tasks, and the preconditions and postconditions for each of the tasks involved in the production of a software.

When software engineers are left to perform tasks based on their own experience, background, and values, they do not necessarily perceive and perform the tasks in the same way or in the same order. They sometimes do not even perform the same tasks. This inconsistency causes projects to take a longer time with poor end products and, in worse situations, total project failure.

## Process defined

- A set of tasks that need to be performed
- The input to and output from each task
- The preconditions and postconditions for each task
- The sequence and flow of these tasks

A software process specification is composed of two basic parts:

- The activities to be included in the software project
- The order in which these activities should be performed

These two main components are further expanded and refined to include the following set of items:

- Activities: Detailed descriptions of each of the activities included in the process
- Control: Necessary entry and exit criteria for each activity, in addition to the order in which each should be performed
- Artifacts: The resulting output from each of the activities
- Resources: The people who perform the activities
- Tools: The tools that may be used to enhance the performance of the activity

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## Process defined

- *A software process is needed to guide and coordinate the group of people.*
- *A software process is a set of related activities that leads to the production of the software. These activities may involve the development of the software from the scratch, or, modifying an existing system.*
- A software process model is a simplified representation of a software process. Each model represents a process from a specific perspective:
  - Waterfall
  - Scrum
  - Lead
  - …

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## Process defined

- Software Process only for a a group of people?
  - What about one-man projects?
    - Documentation
      - *Test Cases*
      - *User guide*
      - *Design Document*
    - Rationale behind decision made
    - Team growth
    - Share information with Stakeholders

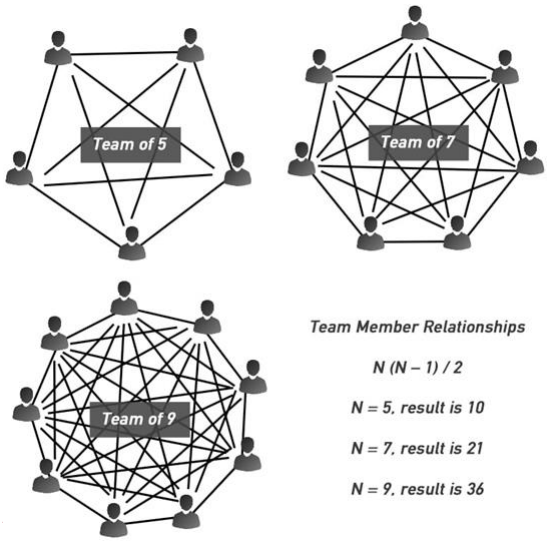
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## Why we need a Process defined?

- Communication
- Commitments
  - Deliverys
- Quality
- Management
  - Product/project progress
- Team morale
  - Ownership
  - Roadmap

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## Process defined – Communication




*Team Member Relationships*

*N (N – 1) / 2*

*N = 5, result is 10*

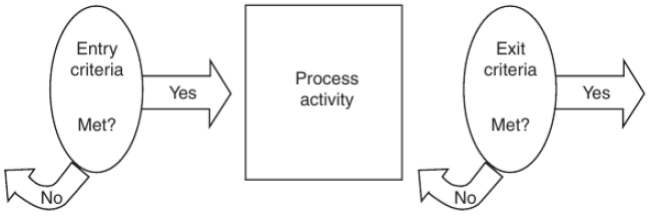
*N = 7, result is 21*

*N = 9, result is 36*

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## Entry and Exit Criteria

- What are the exit criteria for each activity?
- What is the entry criteria for the next follow-on activity?




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## Entry Criteria Examples

- Required artifacts
- Required people
- Required tools
- Required definition of the activity to be performed



## Exit Criteria Examples

- All the artifacts are reviewed
- All or some specified percentage of the errors are corrected
- People in the downstream activities have concurred and accepted the artifacts
- Required definition of the activity to be performed.

- Other criteria may exists:
  - E.g. the person that will participate in the next downstream activity is available.

## Process Assessment Models

- Assessing the maturity of the organization in their software development and support.
- SEI (Software Engineering Institute)
- ISO (International Standards Organization)



## Process Assessment Models

- A process is a set of interrelated or interacting activities which transforms inputs into outputs.

  ISO 9000:2005
- A process model defines:
  - A set of tasks that need to be performed, and its sequence
  - The inputs and outputs from each task
  - Entry and exit criteria for each task
- A lifecycle is the evolution of a system, product, service, project or other human-made entity from conception through retirement

  ISO/IEC 12207
- The process of software development is often modeled as a series of stages that define the software life cycle

  NASA

## ETVX model

- Model introduced by IBM for documenting their processes
  - 'E' - entry criteria which must be satisfied before a set of tasks can be performed
  - 'T' - the set of tasks to be performed
  - 'V' - verification & validation process to ensure that the right tasks are performed
  - 'X' - exit criteria or the outputs of the tasks




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## Case Study - Eating at SELEX



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## Case Study - Eating at SELEX mensa

1. Choose colors for first dish, main dish, and side dish
2. Pick up disk for signaling the main dish
3. Pick up tray with corresponding color for first dish
4. Pick up cup with corresponding color for side dish
5. Choose bread
6. Choose water
7. Choose fruit or desert
8. Put tray on the conveyor
9. Wait to be served
10. Pick up tray from the conveyor
11. Seat
12. Eat
13. Put tray on the other conveyor
14. Separate plastic into bin
15. Exit













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## Eating at SELEX cantina – Process model

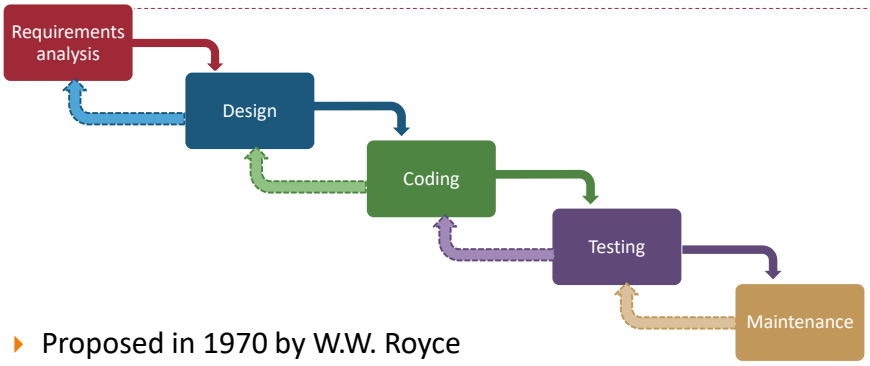
- Lifecycle (sequence of tasks)

Choose menu → Collect food → Eat → Exit

- Process task description: **Choose menu**
  - Entry criteria:
    - Being hungry
    - Be a company employee, or having been invited
  - Inputs:
    - SELEX employee badge or SELEX lunch voucher
  - Activities description:
    - Consult table with menu and corresponding colors
    - Pick up disk with the color of the chosen main dish
    - …
  - Exit criteria:
    - Menu has been chosen
  - Outputs:
    - Tray
    - Disk
    - Cup, knife, fork, spoon

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## Waterfall lifecycle




- Proposed in 1970 by W.W. Royce
- Models a systematic sequential approach – tasks occur sequentially one after another
  - But real projects rarely follow the sequential flow
  - When something goes wrong, go back to a previous step (expensive)

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## Waterfall pros and cons

- Requirements must be specified in the first step
  - Difficult to accommodate the natural uncertainty at the beginning of the projects
  - Sometimes clients demand for closed requirements
- The result of each phase is usually one or more approved documents
  - Document-driven approach – costly iterations
- The output from each stage is fed into the next stage in sequence
- The software project may be tracked as it moves sequentially through specific and identifiable stages.
- Gave SW engineering a process that development could focus.

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## Waterfall pros and cons

- Client sees nothing until the last step
- Works better when requirements are well understood, and technologies known
- Still used as context for software processes

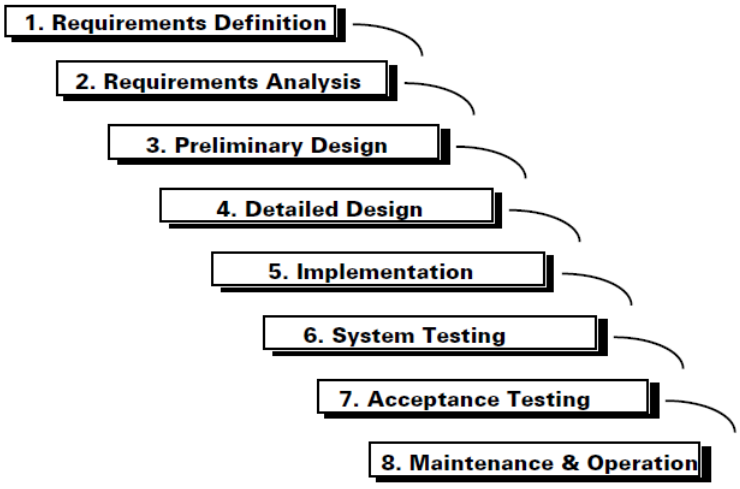


## Classic software life cycle model activities

*Walt Scacchi*

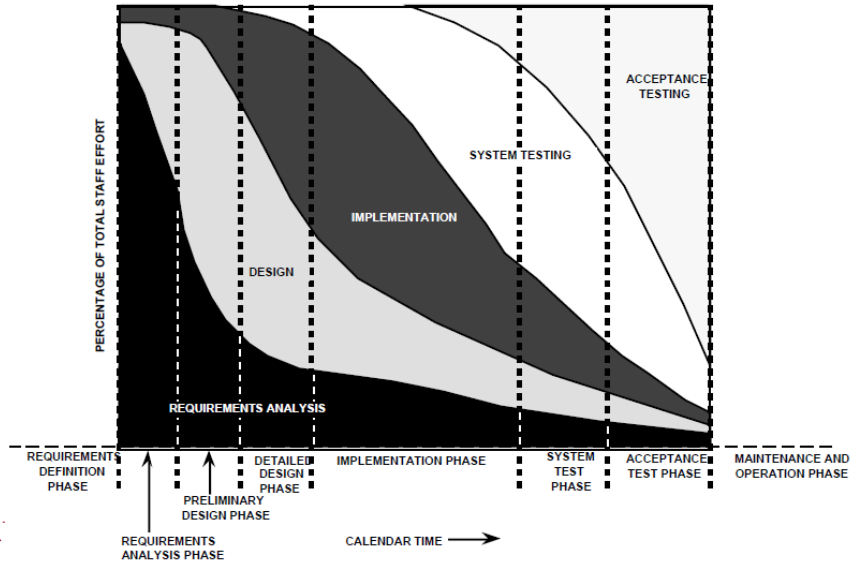
- *System Initiation/Planning*
- *Requirement Analysis and Specification*
- *Functional Specification or Prototyping*
- *Partition and Selection*
- *Architectural Design and Configuration Specification*
- *Detailed Component Design Specification*
- *Component Implementation and Debugging*
- *Software Integration and Testing*
- *Documentation Revision and System Delivery*
- *Deployment and Installation*
- *Training and Use*
- *Software Maintenance*

## NASA-Software Engineering Laboratory recommended software development life cycle

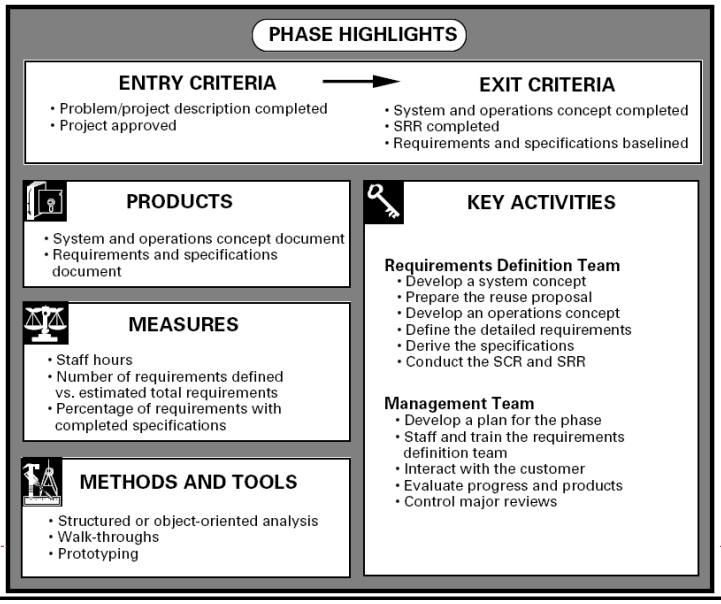



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## NASA-SEL – Activities by Staff Effort



## NASA-SEL ex: REQUIREMENTS DEFINITION PHASE




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## NASA-SEL: Tailoring

- Certain projects, however, do not fit this mold.[…] the life cycle and the methodologies they incorporate often need adjustment.
- Tailoring allows variations in the level of detail and degree of formality of documentation and reviews, which may be modified, replaced, or combined in the tailoring process.
- Such tailoring provides a more exact match to unique project requirements and development products at a lower overall cost to the project without sacrificing quality.

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## ISO/IEC 12207

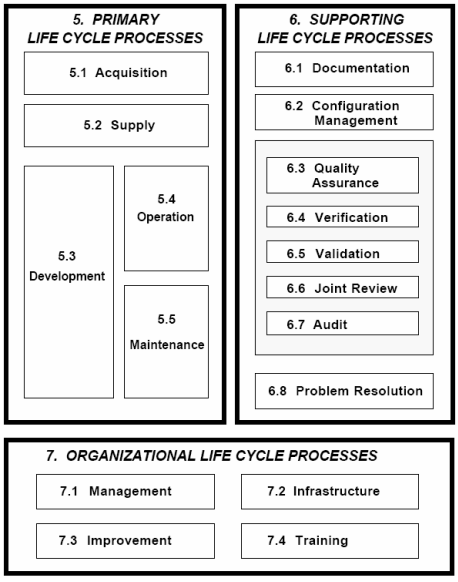
- Information technology- Software life cycle processes (1995)
- Systems and software engineering - Software life cycle processes (2008)
- Systems and software engineering - Software life cycle processes (2017)



## ISO/IEC 12207

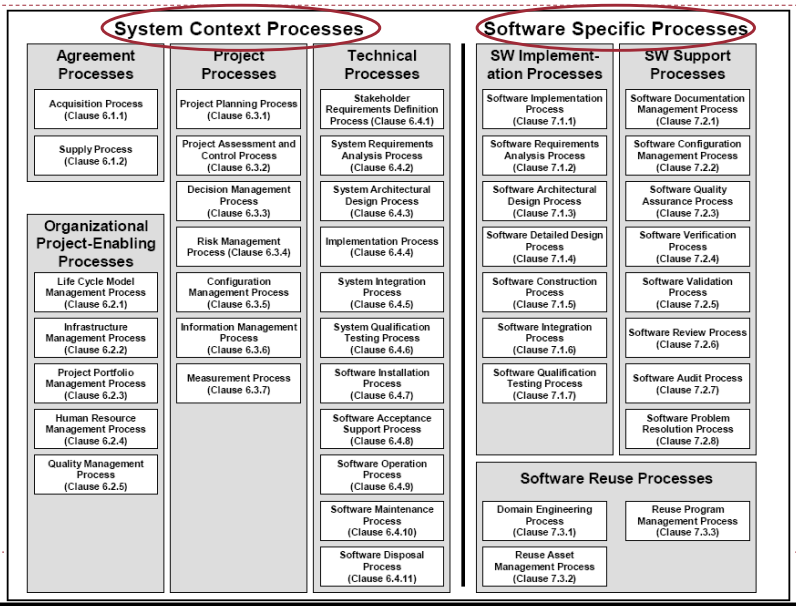
- *Establishes a common framework for software life cycle processes, with well defined terminology, that can be referenced by the software industry. It contains **processes**, **activities**, and **tasks** that are to be applied during the acquisition of a software product or service and during the supply, development, operation, maintenance and disposal of software products.*
- *Does not detail the life cycle processes in terms of methods or procedures […] does not detail documentation in terms of name, format, explicit content and recording media […] does not prescribe a specific system or software life cycle model, development methodology, method, model or technique.*
- Defines stages (phases) and processes. A stage is typically a period of time and ends with a *"primary decision gate"*. The same process often recurs within different stages.

# ISO/IEC 12207:1995




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# ISO/IEC 12207:2008




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## ISO/IEC 12207:2008

- Each process of this standard is described in terms of the following attributes:
  - The title conveys the scope of the process as a whole
  - The purpose describes the goals of performing the process
  - The outcomes express the observable results expected from the successful performance of the process
  - The **activities** are a list of actions that are used to achieve the outcomes
  - The **tasks** are requirements, recommendations, or permissible actions intended to support the achievement of the outcomes.

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## ISO/IEC 12207:2008

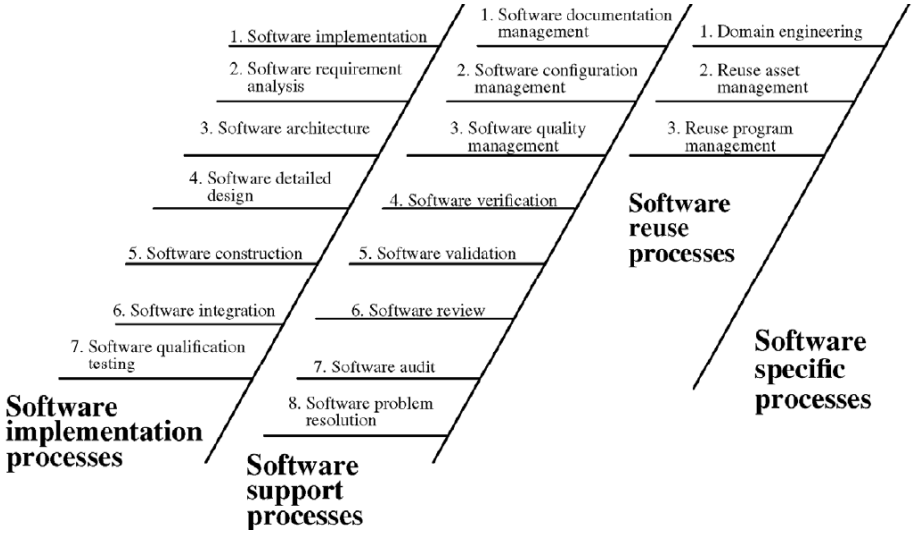
- Within this International Standard clauses numbering:
  - 6.**a** and 7.**a** denote a **process group**
  - 6.a.**b** and 7.a.**b** denote a **process** (or lower-level process) within that group
  - 6.a.b.**1** and 7.a.b.**1** describe the **purpose of the process**
  - 6.a.b.**2** and 7.a.b.**2** describe the **outcome of the process**
  - 6.a.b.**3.c** and 7.a.b.**3.c** list **activities** of the process and clauses
  - 6.a.b.**3.c.d** and 7.a.b.**3.c.d** list **tasks** of the activity 'c'

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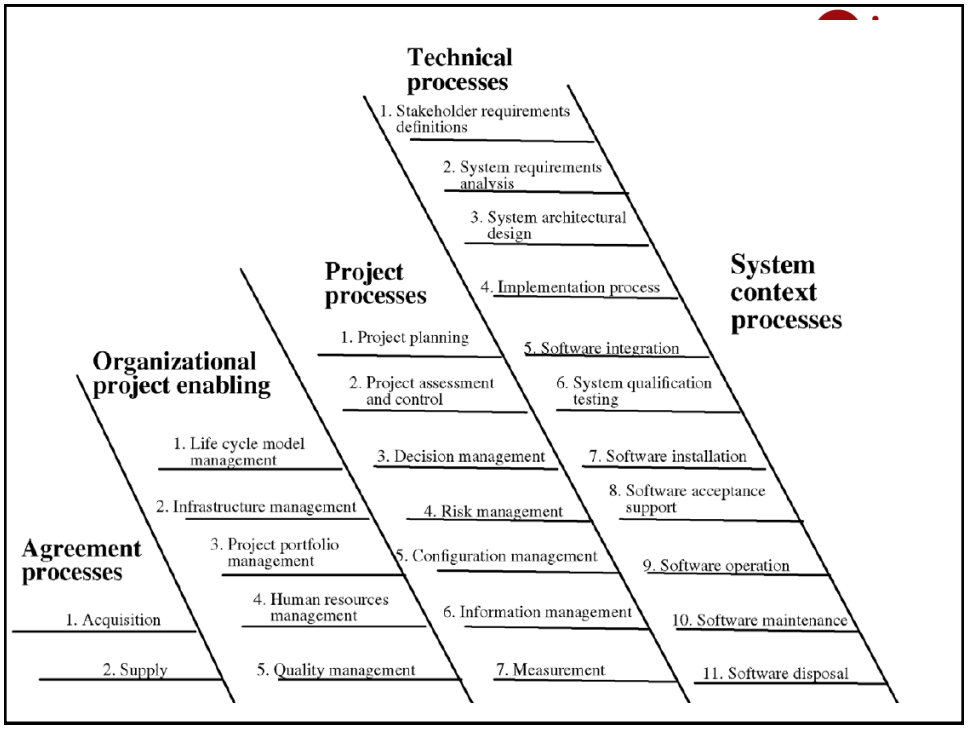
## ISO/IEC 12207:2008 (example)

- 7.1 Software Implementation Processes (**process group**)
- 7.1.2 Software Requirements Analysis Process (**process**)
- 7.1.2.1 Purpose (**purpose of the process**)
  - *The purpose of Software Requirements Analysis Process is to establish the requirements of the software elements of the system.*
- 7.1.2.2 Outcomes (**outcome of the process**)
  - *a) the requirements allocated to the software elements of the system and their interfaces are defined;*
  - *b) software requirements are analyzed for correctness and testability;[…]*
- 7.1.2.3 Activities and tasks
- 7.1.2.3.1 Software requirements analysis (**activity**)
- 7.1.2.3.1.1 The implementer shall establish and document software requirements (including the quality characteristics specifications) described below (**task**).
  - *a) Functional and capability specifications, including performance, physical characteristics, and environmental conditions under which the software item is to perform.*
  - *b) Interfaces external to the software item.*

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**7 Software Specific Processes**

**7.1 Software Implementation Processes**

**7.1.2 Software Requirements Analysis Process**

NOTE The Software Requirements Analysis Process in this International Standard is a lower-level process of the Software Implementation Process. Users of ISO/IEC 15288 may decide that this process is to be provided by the Requirements Analysis Process of ISO/IEC 15288 in a recursive application of that standard.

**7.1.2.1 Purpose**

The purpose of Software Requirements Analysis Process is to establish the requirements of the software elements of the system.

**7.1.2.2 Outcomes**

As a result of successful implementation of the Software Requirements Analysis Process:

a) the requirements allocated to the software elements of the system and their interfaces are defined;

b) software requirements are analyzed for correctness and testability;

**7.1.2.3 Activities and tasks**

The project shall implement the following activities and tasks in accordance with applicable organization policies and procedures with respect to the Software Requirements Analysis Process.

**7.1.2.3.1 Software requirements analysis.** For each software item (or configuration item, if identified) this activity consists of the following tasks:

**7.1.2.3.1.1** The implementer shall establish and document software requirements (including the quality characteristics specifications) described below.

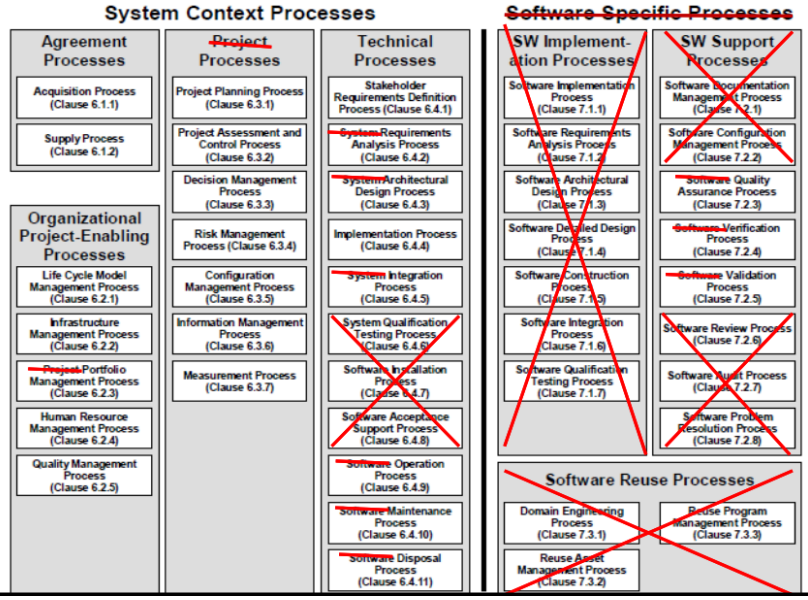
a) Functional and capability specifications, including performance, physical characteristics, and

## ISO/IEC 12207:2008 (tailoring)

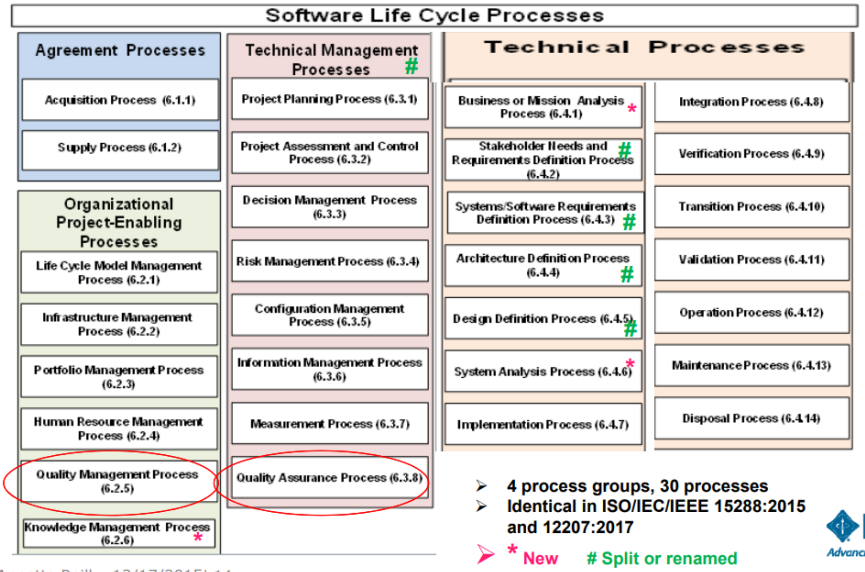
- The purpose of the Tailoring Process is to adapt the processes of this International Standard to satisfy particular circumstances or factors that:
  a) Surround an organization that is employing this International Standard in an agreement.
  b) Influence a project that is required to meet an agreement in which this International Standard is referenced.
  c) Reflect the needs of an organization in order to supply products or services.

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## ISO/IEC 12207:2017



## ISO/IEC/IEEE 12207:2017






## Bibliography


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